THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 1: **Coordinates** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 70	Area $= \frac{1}{2} \times a \times b$.
2	**(a)** 5	The circumcentre is the midpoint of the hypotenuse; its distance from the right-angle vertex is half the hypotenuse.
3	**(b)** 41	Radius $= \frac{1}{2} \times$ distance between the endpoints.
4	**(b)** $(0, 2)$	Set $PA = PB$ with $P = (0, y)$.
5	**(b)** 4	Solve $(x - x_0)^2 + b^2 = c^2$ and take the larger root.
6	**(a)** $(3, 4)$	$D = A + C - B$.
7	**(d)** 3	Solve $(x - a_0)^2 + b^2 = c^2$.
8	**(d)** isosceles right-angled	Reason with coordinates.
9	**(a)** Both A and R are true and R is the correct explanation of A.	$x = -3 < 0$ and $y = 4 > 0$, which is exactly the sign pattern of Quadrant II.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	A point with abscissa 0 is on the y-axis, so A is true; R is also true but is a statement about the x-axis, so it does not explain A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Circumcentre is the midpoint of $AB = (3, 4)$ (1); Radius $= OM = \sqrt{9 + 16} = 5$ (1).	1+1
12	Centre $= (7, 9)$ (1); $AB = 2\sqrt{36 + 64} = 20$, radius $= 10$ (1).	1+1
13	$d^2 = (9)^2 + (-12)^2 = 225$ (1); $d = 15$ (1).	1+1
14	Circumcentre is the midpoint of $AB = (5, 12)$ (1); Radius $= OM = \sqrt{25 + 144} = 13$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Let the point be $(x, 0)$: $(x - 2)^2 + 400 = 625$ (1); $(x - 2)^2 = 225$ (1); $x = 17$ or $x = -13$ (1).	1+1+1
16	$C = B + D - A = (-7, 24)$ (1); $AB = 15$, $AD = 20$ (1); $AC = \sqrt{225 + 400} = 25$ (1).	1+1+1
17	$D = (21, 28)$ (1); $AD = \sqrt{441 + 784}$ (1); $AD = 35$ units (1).	1+1+1
	OR: Let the point be $(x, 0)$: $(x + 1)^2 + 576 = 676$ (1); $(x + 1)^2 = 100$ (1); $x = 9$ or $x = -11$ (1).	1+1+1
18	$C = B + D - A = (1, 27)$ (1); $AB = 20$, $AD = 15$ (1); $AC = \sqrt{400 + 225} = 25$ (1).	1+1+1
	OR: $D = (24, 32)$ (1); $AD = \sqrt{576 + 1024}$ (1); $AD = 40$ units (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $x < 0$, $y > 0$, so Quadrant II (1). (ii) $(-3, -4)$ (1). (iii) Same abscissa, so $AB = \lvert 4 - (-2) \rvert = 6$ km (1). (iv) Midpoint $= \left(\frac{-3 + 5}{2}, \frac{4 - 2}{2} \right) = (1, 1)$ (1); $AC = \sqrt{8^2 + 6^2} = 10$ km (1).	1+1+1+2
20	$AB^2 = BC^2 = CD^2 = DA^2 = 169$ (1); Side $= 13$ (1); $AC^2 = (-7)^2 + (17)^2 = 338$ (1); $AC^2 = BD^2 = 338$: equal diagonals (1); Area $= 13^2 = 169$ square units (1).	1+1+1+1+1
	OR: $PA^2 = PB^2$ with $P = (x, 0)$ (1); $x = -5$, so $P = (-5, 0)$ (1); $PA^2 = 25 + 144 = 169$ (1); $PA = 13$ units (2).	1+1+1+2

Mistakes to avoid:

- Writing the point as (y, x): the abscissa x always comes first, so $(2, 3) \neq (3, 2)$.
- Mixing up the signs of the quadrants, e.g. placing $(-3, 4)$ in Quadrant IV; Quadrant II is $(-, +)$.
- Reflecting in the x-axis but changing the sign of the abscissa (that is a reflection in the y-axis).
- Saying that $(0, -6)$ or $(5, 0)$ lies in a quadrant; points on the axes belong to no quadrant.
- In the distance formula, forgetting the square root or adding the coordinates instead of subtracting them.
- In the midpoint formula, dividing only one coordinate by 2, or subtracting instead of adding.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 1: **Coordinates** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 37	Radius $= \frac{1}{2}\times$ distance between the endpoints.
2	**(c)** $(0, -3)$	Set $PA = PB$ with $P = (0, y)$.
3	**(c)** 3	Solve $(x - x_0)^2 + b^2 = c^2$ and take the larger root.
4	**(a)** $(1, 6)$	$D = A + C - B$.
5	**(a)** 10	Solve $(x - a_0)^2 + b^2 = c^2$.
6	**(a)** collinear, on a vertical line	Reason with coordinates.
7	**(c)** 39	Area $= \frac{1}{2} \times a \times b$.
8	**(d)** 17	The circumcentre is the midpoint of the hypotenuse; its distance from the right-angle vertex is half the hypotenuse.
9	**(c)** A is true but R is false.	$x > 0,\ y < 0$ gives Quadrant IV, so A is true. In Quadrant IV the abscissa is positive, so R is false.
10	**(d)** A is false but R is true.	$(2, 3)$ and $(3, 2)$ are different points, so A is false; R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Centre $= (5, 14)$ (1); $AB = 2\sqrt{81 + 144} = 30$, radius $= 15$ (1).	1+1
12	$d^2 = (-6)^2 + (11)^2 = 157$ (1); $d = \sqrt{157}$ (1).	1+1
13	Circumcentre is the midpoint of $AB = (8, 15)$ (1); Radius $= OM = \sqrt{64 + 225} = 17$ (1).	1+1
14	Centre $= (20, 16)$ (1); $AB = 2\sqrt{400 + 441} = 58$, radius $= 29$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Let the point be $(x, 0)$: $(x - 4)^2 + 1600 = 1681$ (1); $(x - 4)^2 = 81$ (1); $x = 13$ or $x = -5$ (1).	1+1+1
16	$C = B + D - A = (-16, 49)$ (1); $AB = 30$, $AD = 40$ (1); $AC = \sqrt{900 + 1600} = 50$ (1).	1+1+1
17	$D = (11, 60)$ (1); $AD = \sqrt{121 + 3600}$ (1); $AD = 61$ units (1).	1+1+1
	OR: Let the point be $(x, 0)$: $(x + 0)^2 + 1225 = 1369$ (1); $(x + 0)^2 = 144$ (1); $x = 12$ or $x = -12$ (1).	1+1+1
18	$C = B + D - A = (-31, 53)$ (1); $AB = 25$, $AD = 60$ (1); $AC = \sqrt{625 + 3600} = 65$ (1).	1+1+1
	OR: $D = (14, 48)$ (1); $AD = \sqrt{196 + 2304}$ (1); $AD = 50$ units (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $x > 0,\ y < 0$, so Quadrant IV (1). (ii) $(-4, -3)$ (1). (iii) $(4, 3)$ (1). (iv) $F' = (-4, 3)$ (1); $FF' = \sqrt{8^2 + 6^2} = 10$ units (1).	1+1+1+2
20	$AB^2 = BC^2 = CD^2 = DA^2 = 625$ (1); Side $= 25$ (1); $AC^2 = (-17)^2 + (31)^2 = 1250$ (1); $AC^2 = BD^2 = 1250$: equal diagonals (1); Area $= 25^2 = 625$ square units (1).	1+1+1+1+1
	OR: $PA^2 = PB^2$ with $P = (x, 0)$ (1); $x = -8$, so $P = (-8, 0)$ (1); $PA^2 = 49 + 576 = 625$ (1); $PA = 25$ units (2).	1+1+1+2

Mistakes to avoid:

- Writing the point as (y, x): the abscissa x always comes first, so $(2, 3) \neq (3, 2)$.
- Mixing up the signs of the quadrants, e.g. placing $(-3, 4)$ in Quadrant IV; Quadrant II is $(-, +)$.
- Reflecting in the x-axis but changing the sign of the abscissa (that is a reflection in the y-axis).
- Saying that $(0, -6)$ or $(5, 0)$ lies in a quadrant; points on the axes belong to no quadrant.
- In the distance formula, forgetting the square root or adding the coordinates instead of subtracting them.
- In the midpoint formula, dividing only one coordinate by 2, or subtracting instead of adding.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 1: **Coordinates** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 11	Solve $(x-x_0)^2+b^2=c^2$ and take the larger root.
2	**(c)** $(2,3)$	$D=A+C-B$.
3	**(d)** 20	Solve $(x-a_0)^2+b^2=c^2$.
4	**(d)** horizontal line	Reason with coordinates.
5	**(a)** 84	Area $=\frac{1}{2}\times a\times b$.
6	**(a)** 25	The circumcentre is the midpoint of the hypotenuse; its distance from the right-angle vertex is half the hypotenuse.
7	**(c)** 34	Radius $=\frac{1}{2}\times$ distance between the endpoints.
8	**(d)** $(0,-1)$	Set $PA=PB$ with $P=(0,y)$.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Reflection in the x-axis: $(x,y)\to(x,-y)$, which gives $(3,5)$.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	$\sqrt{3^2+4^2}=5$, so A is true; R is true but does not give the value 5.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$d^2=(9)^2+(12)^2=225$ (1); $d=15$ (1).	1+1
12	Circumcentre is the midpoint of $AB=(7,24)$ (1); Radius $=OM=\sqrt{49+576}=25$ (1).	1+1
13	Centre $=(15,13)$ (1); $AB=2\sqrt{144+256}=40$, radius $=20$ (1).	1+1
14	$d^2=(-16)^2+(-12)^2=400$ (1); $d=20$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Let the point be $(x,0)$: $(x+5)^2+576=900$ (1); $(x+5)^2=324$ (1); $x=13$ or $x=-23$ (1).	1+1+1
16	$C=B+D-A=(0,54)$ (1); $AB=40$, $AD=30$ (1); $AC=\sqrt{1600+900}=50$ (1).	1+1+1
17	$D=(27,36)$ (1); $AD=\sqrt{729+1296}$ (1); $AD=45$ units (1).	1+1+1
	OR: Let the point be $(x,0)$: $(x-3)^2+900=1156$ (1); $(x-3)^2=256$ (1); $x=19$ or $x=-13$ (1).	1+1+1
18	$C=B+D-A=(15,62)$ (1); $AB=60$, $AD=25$ (1); $AC=\sqrt{3600+625}=65$ (1).	1+1+1
	OR: $D=(30,40)$ (1); $AD=\sqrt{900+1600}$ (1); $AD=50$ units (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $PQ=8$ m (1). (ii) $\left(\frac{0+8}{2},\frac{0+6}{2}\right)=(4,3)$ (1). (iii) $PR=\sqrt{8^2+6^2}=10$ m (1). (iv) $QS=\sqrt{(0-8)^2+(6-0)^2}=10$ m, equal to PR (1); midpoint of $QS=\left(\frac{8+0}{2},\frac{0+6}{2}\right)=(4,3)$, the same point (1).	1+1+1+2
20	$AB^2=BC^2=CD^2=DA^2=225$ (1); Side $=15$ (1); $AC^2=(-3)^2+(21)^2=450$ (1); $AC^2=BD^2=450$: equal diagonals (1); Area $=15^2=225$ square units (1).	1+1+1+1+1
	OR: $PA^2=PB^2$ with $P=(x,0)$ (1); $x=-3$, so $P=(-3,0)$ (1); $PA^2=144+25=169$ (1); $PA=13$ units (2).	1+1+1+2

Mistakes to avoid:

- Writing the point as (y,x): the abscissa x always comes first, so $(2,3)\neq(3,2)$.
- Mixing up the signs of the quadrants, e.g. placing $(-3,4)$ in Quadrant IV; Quadrant II is $(-,+)$.
- Reflecting in the x-axis but changing the sign of the abscissa (that is a reflection in the y-axis).
- Saying that $(0,-6)$ or $(5,0)$ lies in a quadrant; points on the axes belong to no quadrant.
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